

Research paper

Optimal control scheme for multi-terminal high-voltage direct current system to improve robust grid frequency regulations

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ABSTRACT

This paper proposes an optimal control scheme for robust frequency regulation (FR) in power networks that utilize multi-terminal high-voltage direct current (MT-HVDC) systems to interconnect AC grids. The proposed scheme involves the development of a Linear Quadratic Gaussian (LQG) controller to minimize FR in each grid. A state space model of the MT-HVDC grids is derived to design the controller, considering the DC-link voltage and active power controls at the rectifier and inverter, respectively. Additionally, the model incorporates inertia emulation (IE) and droop controllers. Based on the derived state-space model, an LQG controller is designed to achieve optimal frequency control. The stability of the proposed scheme is analyzed through sensitivity and eigenvalue analysis, with a focus on the influence of grid parameters and control gains. Comparative case studies demonstrate that the proposed scheme outperforms conventional strategies in terms of improving the effectiveness and robustness of real-time FR in MT-HVDC grids. The proposed control scheme involves an LQG controller that considers various factors and demonstrates superior performance compared to conventional methods, as supported by comparative case studies.

1. Introduction

Contemporary power grids have witnessed notable improvements in system reliability and efficiency with the increased incorporation of high-voltage direct current (HVDC) systems (Sessa et al., 2019). Among these systems, VSC-based HVDC technology has gained popularity owing to its advantages, including asynchronous grid connection, rapid response of power converters to reference signals, and independent control of active and reactive power (Schonleber et al., 2020). Recently, VSC-based MT-HVDC systems have emerged as feasible solutions for transmitting substantial power from large-scale renewable energy sources (RESs) or the mainland to relatively smaller power grids, such as island grids (Raza et al., 2018). A prime example is the Zhangbei HVDC system, comprising a symmetrical four-terminal bipolar HVDC system, which has demonstrated significant enhancements in grid reliability. In this project, all four stations were connected to their respective local AC grids. The system operates at a nominal DC voltage of ± 500 kV, with a nominal power rating of 3000 MW for the Changping station and 1500 MW for the remaining three stations (Wang et al., 2019; Gao et al.,

2020; Li et al., 2021). Furthermore, VSC-based MT-HVDC systems enable FR for power grids connected to the mainland by adjusting the transfer power (Lin and Wu, 2021).

Several studies (Zhu et al., 2013; Shen et al., 2022; Zhu et al., 2022; Wang et al., 2023) have investigated real-time FR of VSC-based HVDC systems to reduce grid-frequency fluctuations. Zhu et al. (2013) proposed a control strategy for emulating inertia in VSC-based HVDC systems by utilizing the energy stored in DC-link capacitors. A previous study (Shen et al., 2022) focused on the FR of VSC-HVDC systems by considering a variable-inertia emulation control (VIEC) scheme. Zhu et al. (2022) proposed an inertia emulation and fast frequency-droop (IEFF) control strategy, in which the power-regulating VSC at one end emulates a SG via active power control. Wang et al. (2023) introduced a multiple adaptive model predictive controller (MAMPC) that integrates the concepts of multiple model predictive control (MMPC) and a real-time autoregressive with exogenous input (ARX) model to minimize the frequency deviation. However, these studies only focused on supporting FR in single-terminal networks. Therefore, additional strategies are required to address FR in an MT-HVDC system.

Different approaches have been proposed to address FR in MT-HVDC

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Nomenclature

A. Acronyms

FR	Frequency regulation
HVDC	High-voltage direct-current
IE	Inertia emulation
LQR	Linear Quadratic Regulator
LQG	Linear Quadratic Gaussian
PCC	Point of common coupling
PFC	Primary frequency control
PI	Proportional-integrated
RVSC	Rectifier side VSC
SFC	Secondary frequency control
SG	Synchronous generator
VSC	Voltage source converter

B. Sets

r, n	Rectifier and Numbers of inverters
g, l	Subscripts for the variables of the grid and load in the MT-HVDC grids
$max, rated, ref, rms$	Subscripts for the maximum, rated, reference, and root-mean-square values

C. MT-HVDC system

$v_{sr}, v_{sn}, i_{sr}, i_{sn}$	Voltages and currents of the HVDC rectifier and inverter at the PCC
$v_{cr}, v_{cn}, i_{cr}, i_{cn}$	Voltages and currents of the HVDC rectifier and inverter at the VSC terminals
V_{dcr}, V_{dcn}	DC-link voltages of the rectifier and inverter
I_{dcr}, I_{dcn}	DC-link currents of the rectifier and inverter
P_r, P_n	Active power at the rectifier and inverter
Q_r, Q_n	Reactive power at the rectifier and inverter
C_{dcr}, C_{dcn}	DC-link capacitance of the rectifier and inverter
R_{eq}, L_{eq}	Total resistance and inductance between the VSC terminals and the PCC
R_{dc}, L_{dc}	Total resistance and inductance of the DC-link
z_{1-4r}, z_{1-4n}	Integral state variables of the outer and inner loops at the rectifier and inverter

$k_{p, vdc}, k_{i, vdc}$ The proportional and integral gains of the outer loops at the rectifier

$k_{p, pq}, k_{i, pq}$ The proportional and integral gains of the outer loops at the inverter

$k_{p, dq}, k_{i, dq}$ The proportional and integral gains of the inner loops at the rectifier and the inverter

D. MT-HVDC grids

P_{gn}, P_{ln}	Active power from the total synchronous generator and the load demands at the grids
f_n	Frequency at the grids
M, D	Inertia and damping of the grid
T_g, T_{ch}	Time constants of the generator
T_{fn}, v_n	Time constants and coefficients for IE
R_g, R_n	Droop gains of the generators and grids
t_d	Communication time delay

E. State-space model

A_v, B_v, C_v	Coefficient matrices of a state-space model of the MT-HVDC system
X_v, U_v, Y_v	States, inputs, and outputs of a state-space model of the MT-HVDC system
A_G, B_G, C_G	Coefficient matrices of a state-space model of the MT-HVDC linked grids
X_G, U_G, Y_G	States, inputs, and outputs of a state-space model of the MT-HVDC linked grids
A_{td}, B_{td}, C_{td}	Coefficient matrices of a state-space model of the MT-HVDC linked grids considering time delays
X_{td}, U_{td}, Y_{td}	States, inputs, and outputs of a state-space model of the MT-HVDC linked grids considering time delays
A_R, B_R, C_R	Coefficient matrices of a state-space model of MT-HVDC linked grids including LQG controller
X_R, U_R, Y_R	States, inputs, and outputs of a state-space model of MT-HVDC linked grids including LQG controller
K_r	Optimal LQG gain matrix
Q, R	Diagonal weighting matrices for the state and input matrices

systems (Silva et al., 2012; Kirakosyan et al., 2022; Kirakosyan et al., 2018; Wang et al., 2018). For instance, Silva et al. (2012) introduced a cascading control algorithm based on frequency controls at the inverters to improve the FR in MT-HVDC systems connected to inverters. Kirakosyan et al. (2022) presented a voltage droop control scheme with an additional voltage-shifting term to enhance the mutual frequency support between interconnected MT-HVDC linked grids. Kirakosyan et al. (2018) also implemented VSCs with droop control to address two key challenges in wind power generation: grid frequency deviations and variations in DC voltage. These VSCs utilized droop control to effectively mitigate the aforementioned issues, thereby ensuring the stability and reliability of the overall grid system. Wang et al. (2018) presented a comprehensive analysis of adaptive droop control applied to a VSC-based MT-HVDC system. Their study focused on examining the V_{dc} - I_{dc} - f characteristics of the system to determine the optimal droop-control gain. However, these studies primarily employed droop controllers for PFC or conventional PI controllers to support FR in MT-HVDC systems rather than the optimal SFC. This statement highlights the necessity for additional research aimed at augmenting the performance of FR. Further investigation is warranted to improve the FR efficiency.

Recently, scholars have explored the mitigation of grid-frequency fluctuations using the SFC of MT-HVDC systems. Ambia et al. (2021) proposed an optimization algorithm based on the consensus method

tuning a droop coefficient to minimize generation costs and converter losses in MT-HVDC systems. Wang et al. (2021) presented a distributed cooperative control algorithm utilizing a local photovoltaic (P - V) droop controller and cooperative agent network to minimize grid-frequency deviations. Jia et al. (2020) implemented a distributed economic model predictive control (EMPC) algorithm to achieve optimal power sharing in a VSC-based MT-HVDC system. The study involved formulating a discrete-time optimization problem that considered grid operating costs. Mehrabankhomartash et al. (2020) presented MPC-based power sharing as a solution to address AC line overloads in MT-HVDC systems. The proposed strategy involves periodic updates of the power set-points for both MT-HVDC converter stations and AC generators. Moreover, it should be noted that MPC can be computationally demanding because it requires solving an optimization problem at each time step. This computational intensity arises from the need to repeatedly address the MPC optimization problem directly (Raja et al., 2023). Furthermore, the implementation of MPC in MT-HVDC linked grids can introduce instability because of communication delays. Therefore, further analyses are required to improve the practical applicability of FR in MT-HVDC systems.

This paper proposes a novel real-time frequency control scheme for MT-HVDC grids to improve FR performance by considering the differences between grid characteristics and communication time delays between the proposed and local controllers. A state-space model of the

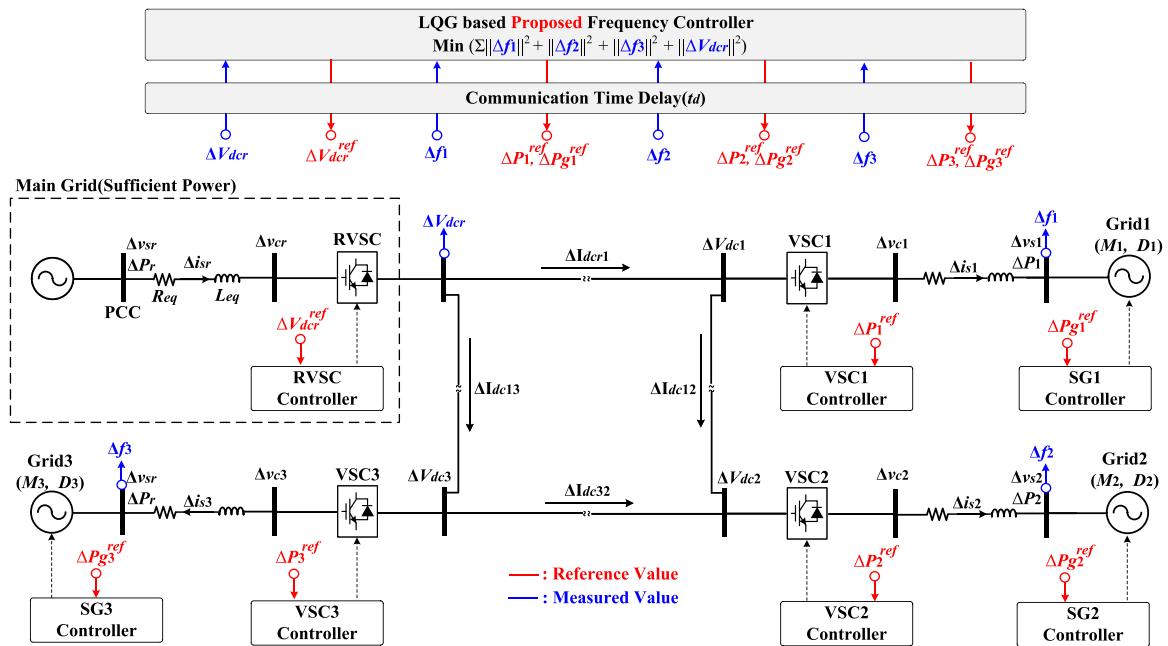


Fig. 1. Schematic of MT-HVDC grids that use the proposed control strategy to perform real-time FR.

MT-HVDC grids was derived and validated by comparing it to a detailed MATLAB/Simulink model. The model served as the basis for designing real-time FR with PFC and optimal SFC to mitigate grid-frequency fluctuations, considering inaccuracies in HVDC system operations and optimal SFC. Sensitivity and eigenvalue analyses were conducted to verify the grid frequency in the proposed control scheme, with a specific focus on examining the impact of system parameters and communication time delays. Moreover, comparative case studies were conducted to assess the effectiveness of the proposed scheme in mitigating frequency deviations across various grid scenarios. The main contributions of this study are summarized below.

- This paper presents the optimal real-time FR of VSC-based MT-HVDC linked grids using the LQG controller. The LQG controller was designed based on measurement data and communication time delays within the grid between the proposed and local controllers.
- A detailed state-space model of the MT-HVDC linked grids was derived to develop an LQG controller to achieve optimal FR by considering communication time delays. The MT-HVDC linked grids were modeled using the control input and output measured under

normal conditions. The derived state-space model successfully captured the dynamic profiles of the grid frequency and power, providing a comprehensive representation of system behavior.

- The proposed control scheme was rigorously validated by comparing it with controllers installed in an actual system, with a particular emphasis on its effectiveness and robustness in mitigating frequency deviations caused by communication delays. An authentic model of an MT-HVDC project was employed in conjunction with conventional test grids to ensure precise validation of the proposed controller. This approach enhanced the accuracy and reliability of controller verification, demonstrating the superior performance of the proposed LQG controller in minimizing frequency deviations and ensuring stable grid operation.

2. Fundamentals and overall structure

Fig. 1 presents the proposed schematic for optimal FR in MT-HVDC linked grids. The diagram depicts four interconnected VSC-based MT-HVDCs. For simplicity, each grid comprises SGs and loads. The frequency variations (Δf_n) in each grid are influenced by various factors,

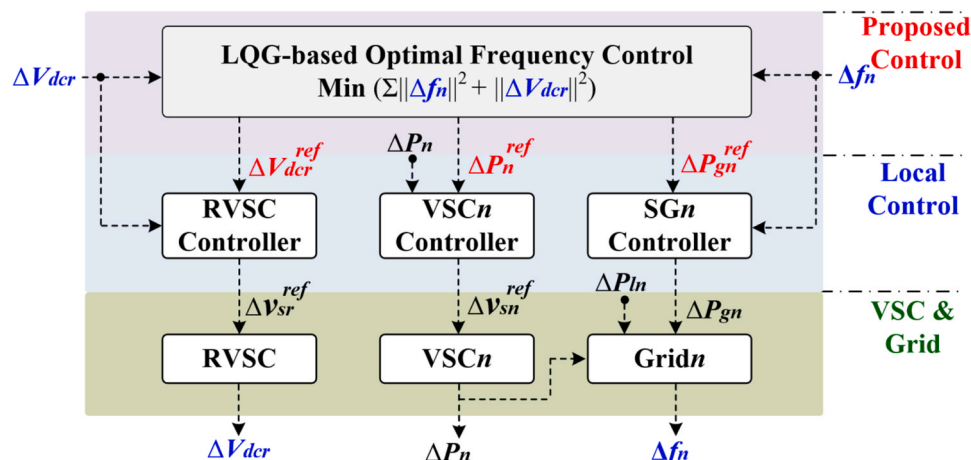


Fig. 2. Proposed control structure for optimizing reference profiles in MT-HVDC.

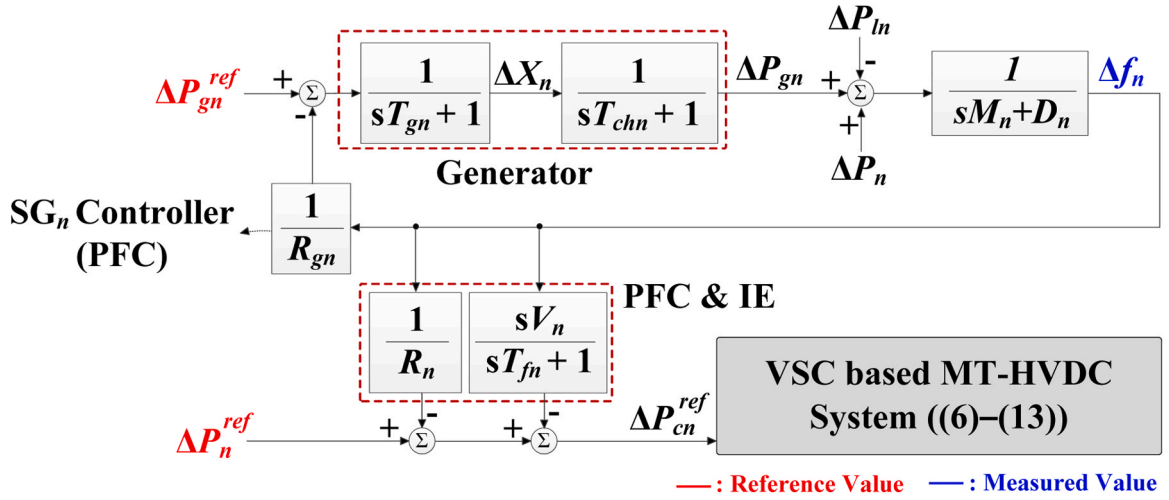


Fig. 3. Block diagram of the MT-HVDC System with PFC and IE schemes.

including load demand variations (ΔP_{ln}), generator power variations (ΔP_{gn}), and intermittent power outputs of the grid-connected VSCs (ΔP_n) (Kim et al., 2020). The main grid connected to the RVSC is assumed to possess sufficient active power to support the FR in other grids (Muyeen et al., 2010). Additionally, in this paper, the study is conducted assuming that the main grid is very large in size, unlike other grids. For example, if the main grid is the mainland, all other grids except the main grid are islands. The proposed control scheme aims to minimize the combined value of the weighted quadratic sum of frequency fluctuation of the grid (Δf_n) and DC voltage variation of RVSC (ΔV_{dcr}). To support optimal FR in the grids, the proposed control scheme determines the reference power signals for the grid side VSCs (VSC_n) and SGs. The proposed controller receives measurement data, such as Δf_n and ΔV_{dcr} , from both the grids and RVSC via a communication system with time delays (Pipelzadeh et al., 2012). As depicted in Fig. 1, the local controller of the RVSC regulates the DC voltage, whereas that of the VSC_n control the active power. Each grid in the system has different inertia (M) and damping (D) characteristics that determine the reliability of the proposed controller based on the LQG control algorithm. In this paper, it is assumed that there is no communication noise or loss between the central controller and VSC local controllers.

Fig. 2 illustrates the application of an LQG control scheme to the proposed control structure to achieve optimal real-time FR in the grids. The proposed controller communicates with the local controllers and transmits the optimal reference signals ΔP_n^{ref} , ΔV_{dcr}^{ref} and ΔP_{gn}^{ref} to each local controller in the VSCs and SGs to regulate frequency and DC voltage, respectively, thereby ensuring stable and reliable grid operation. Specifically, the measured data for Δf_n and ΔV_{dcr} are transmitted to the controller. Local controllers in VSC_n, RVSC, and SGs derive reference signals for voltage at the PCC (ΔV_{sr}^{ref} , ΔV_{sn}^{ref}) and generator power variations (ΔP_{gn}) using the reference value transmitted from the proposed controller within the control loops, including both inner and outer control mechanisms. Therefore, the RVSC and VSC_n control ΔV_{dcr} and ΔP_n , respectively, to track the reference signals (Kalcon et al., 2012).

3. MT-HVDC linked grids considering time delay

An HVDC system comprises a rectifier, inverter, and DC link (Cole et al., 2010; Zhou et al., 2014). The VSC_n can be modeled as

$$L_{eq} \frac{di_{sn}^{dq}}{dt} = -R_{eq} i_{sn}^{dq} \pm \omega L_{eq} i_{sn}^{qd} + v_{sn}^{dq} - v_{cn}^{dq}, n = 1, 2, 3, \quad (1)$$

$$C_{dc} \frac{dV_{dcr}}{dt} = (i_{sn}^d \cdot v_{cn}^d) / V_{dcr} + I_{dcr}, n = 1, 2, 3, \quad (2)$$

In this study, the converters of RVSC and VSC_n were equipped with commonly used PI-based outer and inner controllers to mitigate static

errors in the control system (Ibrahim and Dessouky, 2021). The primary objective of the outer feedback loop was to regulate the V_{dcr} of the RVSC and P_n of the VSC_n. Subsequently, these reference signals were fed into the VSCs to establish the desired current references. The reference current values facilitated the operation of the outer feedback loop. Upon obtaining the reference values, the inner feedback loop was used to determine the V_{sr} and V_{sn} of the VSCs in alignment with the d - q axis. The outer and inner controllers of the VSC_n were modeled as

$$i_{sn}^{dq-ref} = k_{p-PQ} (P_n^{ref} - P_n) + z_n, n = 1, 2, 3, \quad (3)$$

$$v_{cn}^{dq} = v_{sn}^{dq} \omega L_{eq} i_{sn}^{qd} - k_{p-dq} (i_{sn}^{dq-ref} - i_{sn}^{dq}) - m_n, n = 1, 2, 3, \quad (4)$$

where $dz_n/dt = k_{i-PQ} (P_n^{ref} - P_n)$ and $dm_n/dt = k_{i-dq} (i_{sn}^{dq-ref} - i_{sn}^{dq})$. In Eqs. (3) and (4), k_{p-PQ} and k_{i-PQ} denote the proportional and integral gains of the outer control schemes in the VSC-HVDC system, respectively, and k_{p-dq} and k_{i-dq} denote the proportional and integral gains of the inner control schemes, respectively.

The equations related to the DC current and voltage flowing through the DC-link network of the rectifier and inverter are as follows:

$$V_{dcr} - V_{dcrn} = L_{dc} \frac{dI_{dc}}{dt} + R_{dc} I_{dc}, n = 1, 2, 3, \quad (5)$$

Given the VSC models and DC-link network (Eqs. (1)–(5)), the MT-HVDC system can be represented in a state-space form as follows:

$$d\mathbf{X}_v(t)/dt = \mathbf{A}_v \mathbf{X}_v(t) + \mathbf{B}_v \mathbf{U}(t), \quad (6)$$

$$\mathbf{Y}_v(t) = \mathbf{C}_v \mathbf{X}_v(t), \quad (7)$$

Where

$$\mathbf{A}_v = \begin{bmatrix} \mathbf{A}_r & & & \mathbf{0} \\ & \mathbf{A}_1 & & \mathbf{0} \\ & & \mathbf{A}_2 & \mathbf{0} \\ & & & \mathbf{A}_3 & \mathbf{0} \\ & & & & \mathbf{A}_{dc} \end{bmatrix}, \quad (8)$$

$$\mathbf{B}_v(t) = [\mathbf{B}_r(t)^T, \mathbf{B}_1(t)^T, \mathbf{B}_2(t)^T, \mathbf{B}_3(t)^T, \mathbf{0}]^T, \quad (9)$$

$$\mathbf{C}_v(t) = [\mathbf{C}_r(t)^T, \mathbf{C}_1(t)^T, \mathbf{C}_2(t)^T, \mathbf{C}_3(t)^T, \mathbf{0}]^T, \quad (10)$$

$$\mathbf{X}_v(t) = [\Delta i_{sr}^{dq}, \Delta V_{dcr}, \Delta z_n, \Delta i_{sn}^{dq}, \Delta V_{dcrn}, \Delta m_n, \Delta I_{dc}, \Delta P_{cn}^{ref}]^T, n = 1, 2, 3, \quad (11)$$

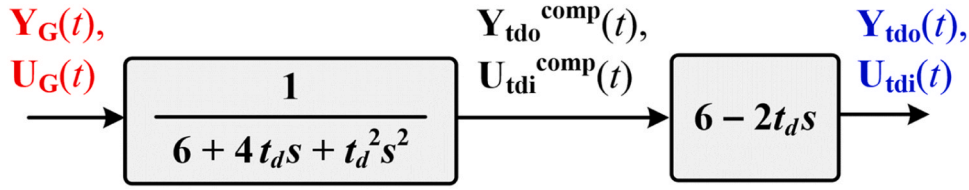


Fig. 4. Padé method to reflect the communication delays in $U_{td}(t)$ and $Y_{td}(t)$.

$$U_v(t) = [\Delta V_{dcr}^{ref}, \Delta P_n^{ref}]^T, \quad n = 1, 2, 3, \quad (12)$$

$$Y_v(t) = [\Delta V_{dcr}, \Delta P_n]^T, \quad n = 1, 2, 3. \quad (13)$$

where A_r , A_{1-3} , B_r , B_{1-3} , and C_r , C_{1-3} are presented in the Appendix. Fig. 3 illustrates a block diagram of the MT-HVDC system with the PFC and IE schemes, where the MT-HVDC system supports real-time FR on grids. The derived state-space model of the MT-HVDC system is also included. The PFC scheme utilizes droop controllers characterized by the coefficient R_n to ensure the stabilization of Δf_n within localized regions. Furthermore, the IE schemes are represented through supplementary controllers, which are strategically designed to enhance the integration of rapid responses into the operation of the VSCs. Considering the PFC and IEs of the MT-HVDC system, a dynamic model of the MT-HVDC linked grids was developed by integrating the individual models of the AC grids, VSCs, and DC network, as discussed in the Appendix.

$$dX_G(t)/dt = A_G \cdot X(t) + B_G \cdot U(t), \quad (14)$$

$$Y_G(t) = C_G \cdot X_G(t), \quad (15)$$

with the states, inputs, and outputs arranged, respectively, as follows:

$$X_G(t) = [X_v(t), \Delta P_{gn}, \Delta f_n, \Delta X_n]^T, \quad n = 1, 2, 3, \quad (16)$$

$$U_G(t) = [U_v(t), \Delta P_{gn}^{ref}, \Delta P_{ln}]^T, \quad n = 1, 2, 3, \quad (17)$$

$$Y_G(t) = [Y_v(t), \Delta P_{gn}, \Delta f_n]^T, \quad n = 1, 2, 3. \quad (18)$$

Generally, time delays are represented using the second-order Padé approximation method. This method provides a reasonable phase approximation but introduces a non-zero response at the origin. In this study, the second-order approximation method yielded a slightly improved representation, particularly for longer time delays (Dotta et al., 2009). The approximation method was employed to model the communication time delays (t_d) between the central and local controllers. Fig. 4 illustrates that $Y_{tdo}^{comp}(t)$ can be represented as follows:

$$Y_{tdo}^{comp}(t) = Y_G(t) \times \frac{1}{6 + 4t_d s + t_d^2 s^2}, \quad (19)$$

where $Y_{tdo}^{comp}(t) = x_{tdo1}(t)$ and $Y_{tdo}^{comp}(t) = x_{tdo1}(t) = x_{tdo2}(t)$. The Laplace transform of (19) is as follows;

$$Y_{tdo}^{comp}(s) = x_{tdo1}(s) = Y_G(s) \times \frac{1}{6 + 4t_d s + t_d^2 s^2}, \quad (20)$$

The equation in (20) can be summarized into the Eqs. (21)–(24) as follows;

$$Y_G(s) = (6 + 4t_d s + t_d^2 s^2) x_{tdo1}(s), \quad (21)$$

$$Y_G(s) = 6x_{tdo1}(s) + 4t_d s x_{tdo1}(s) + t_d^2 s^2 x_{tdo1}(s), \quad (22)$$

$$t_d^2 \dot{x}_{tdo2}(t) = Y_G(t) - 6x_{tdo1}(t) - 4t_d x_{tdo2}(t) \quad (23)$$

$$\dot{x}_{tdo2}(t) = -\frac{6}{t_d^2} x_{tdo1}(t) - \frac{4}{t_d} x_{tdo2}(t) + \frac{1}{t_d^2} Y_G(t) \quad (24)$$

If the equation for (24) is organized into a matrix, it is as follows;

$$\begin{bmatrix} \dot{x}_{tdo1}(t) \\ \dot{x}_{tdo2}(t) \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ -6/t_d^2 & -4/t_d \end{bmatrix}}_{A_{td}} \begin{bmatrix} x_{tdo1}(t) \\ x_{tdo2}(t) \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ 1/t_d^2 \end{bmatrix}}_{B_{td}} Y_G(t), \quad (25)$$

An output signal considering t_d in the MT-HVDC grids $Y_{tdo}(t)$ is then expressed as:

$$Y_{tdo}(t) = Y_{tdo}^{comp}(t) \times (6 - 2t_d s) \quad (26)$$

The equation for (26) can be summarized as follows;

$$\begin{aligned} Y_{tdo}(s) &= X_{tdo1}(s) \times (6 - 2t_d s) \\ &= 6X_{tdo1}(s) - 2t_d s X_{tdo1}(s) \\ &\quad \underbrace{X_{tdo2}(s)} \end{aligned} \quad (27)$$

If the equation for (27) is organized into a matrix, it is as follows;

$$Y_{tdo}(t) = \underbrace{\begin{bmatrix} 6 & 0 \\ 0 & -2t_d \end{bmatrix}}_{C_{td}} \begin{bmatrix} x_{tdo1}(t) \\ x_{tdo2}(t) \end{bmatrix} \quad (28)$$

Similarly, $X_{tdi}(t)$ for $U_G(t)$ is defined as;

$$\begin{bmatrix} \dot{x}_{tdi1}(t) \\ \dot{x}_{tdi2}(t) \end{bmatrix} = A_{td} \begin{bmatrix} x_{tdi1}(t) \\ x_{tdi2}(t) \end{bmatrix} + B_{td} U_G(t), \quad (29)$$

$$U_{tdi}(t) = C_{td} \cdot X_{tdi}(t) \quad (30)$$

The incorporation of delays into the system model can be accomplished by utilizing a state-space representation. One approach involves defining system matrices that consider the input and output delays, as depicted in Fig. 4. The resulting system matrices that consider these delays are generally represented by Eqs. (31) and (32), respectively.

$$dX_{td}(t)/dt = A_{td} \cdot X_{td}(t) + B_{td} \cdot U_{td}(t), \quad (31)$$

$$Y_{td}(t) = C_{td} \cdot X_{td}(t), \quad (32)$$

where

$$A_{td} = \begin{bmatrix} A_G & B_G C_{tdi} & 0 \\ 0 & A_{tdi} & 0 \\ B_{tdo} C_G & 0 & A_{tdo} \end{bmatrix}, B_{td} = \begin{bmatrix} 0 \\ B_{tdi} \\ 0 \end{bmatrix} \quad (33)$$

$$C_{td} = [0, 0, C_{tdo}], \quad (34)$$

$$X_{td}(t) = [X_G(t)^T, X_{tdi}(t)^T, X_{tdo}(t)^T]^T, \quad (35)$$

$$Y_{td}(t) = Y_{tdo}(t). \quad (36)$$

4. Proposed optimal frequency control algorithm

4.1. Modeling of proposed FR applied to the LQG controller

For optimal SFC, the dynamic model of MT-HVDC linked grids was

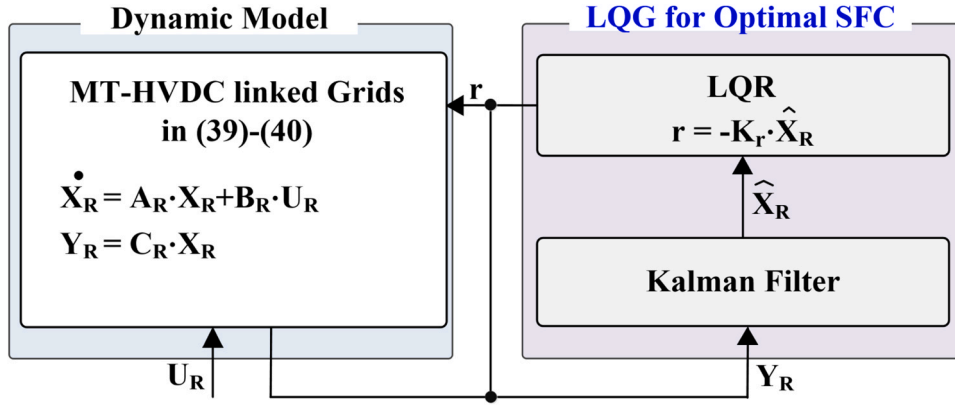


Fig. 5. LQG control scheme integrated with the dynamic model of the MT-HVDC linked grids.

augmented by including Δf_n and ΔV_{dcr} as follows:

$$\mathbf{X}_R(t) = \left[\Delta V_{dcr}, \Delta f_n, \int \Delta V_{dcr} dt, \int \Delta f_n dt, \mathbf{X}_G(t) \right]^T, \quad n = 1, 2, 3, \quad (37)$$

$$\mathbf{U}_R(t) = \left[\Delta V_{dcr}^{ref}, \Delta P_n^{ref}, \Delta P_{gn}^{ref}, \Delta P_{in} \right]^T, \quad n = 1, 2, 3, \quad (38)$$

$$\mathbf{Y}_R(t) = \left[\Delta V_{dcr}, \Delta f_n, \int \Delta V_{dcr} dt, \int \Delta f_n dt, \mathbf{Y}_G(t) \right]^T, \quad n = 1, 2, 3, \quad (39)$$

The augmented state–space model using Eqs. (37)–(39) can be expressed concisely as

$$d\mathbf{X}_R(t)/dt = \mathbf{A}_R \cdot \mathbf{X}_R(t) + \mathbf{B}_R \cdot \mathbf{U}_R(t), \quad (40)$$

$$\mathbf{Y}_R(t) = \mathbf{C}_R \cdot \mathbf{X}_R(t), \quad (41)$$

As shown in Fig. 5, the LQG controller combines the LQR and Kalman filter (Kwon et al., 2020). The LQR, represented by the control input equation $\mathbf{r}(t) = -\mathbf{K} \cdot \mathbf{X}_R(t)$, was integrated with the augmented state–space model described by Eqs. (40) and (41). The integration was aimed at minimizing the cumulative variations in f_n and V_{dcr} , while ensuring that the recovered value equaled the nominal values in steady state. By considering the control efforts of both the MT-HVDC system and generators, the cost function for the proposed scheme can be formulated as follows:

$$J = \int_0^\infty (\mathbf{X}_R(t)^T \cdot \mathbf{Q} \cdot \mathbf{X}_R(t) + \mathbf{r}(t)^T \cdot \mathbf{R} \cdot \mathbf{r}(t)) dt. \quad (42)$$

Eq. (42) represents the equation for $\mathbf{P}$, which is the solution to the optimization problem that minimizes J . In Eq. (42), the matrix $\mathbf{Q}$ represents a diagonal matrix with weighting coefficients. Each coefficient is multiplied by the square of the corresponding state variable, such as $\int \Delta f_n dt$, $\int \Delta V_{dcr} dt$, Δf_n , and ΔV_{dcr} , in $\mathbf{X}_R(t)$. The term $\mathbf{X}_R(t)^T \cdot \mathbf{Q} \cdot \mathbf{X}_R(t)$ corresponds to the weighted sum of the variations accumulated in real time in the frequency of each grid and the DC voltage of the RVSC. By multiplying $\mathbf{X}_R(t)$ with $\mathbf{Q}$ and then determining the inner product with $\mathbf{X}_R(t)$, the weighted sum was obtained, which represents the combined effect of variations in frequency and DC voltage. This term is crucial in the overall cost function and captures the objective of minimizing these variations in the control scheme. Similarly, the matrix $\mathbf{R}$ represents a diagonal matrix with weighting coefficients multiplied by the squares of the input variables in $\mathbf{r}(t)$. The matrix $\mathbf{R}$ enables the term $\mathbf{r}(t)^T \cdot \mathbf{R} \cdot \mathbf{r}(t)$ to be included and thus incorporates the control efforts of the MT-HVDC linked grids to reduce the weighted sum of the frequency variations. The term represents the contribution of the control efforts to the overall cost function, emphasizing the objective of minimizing the weighted sum of the grid frequency variations. The weighting coefficients in $\mathbf{R}$ determine the relative importance of different control inputs in the cost function and

enable the optimization process to prioritize certain control actions over others to achieve the desired control objectives. In particular, the system operator of the MT-HVDC grids can adaptively obtain the coefficients in $\mathbf{Q}$ and $\mathbf{R}$ by considering the operating conditions. For example, if the coefficients of the terms for Δf_1 are larger than those for Δf_2 , Δf_1 can be assigned a smaller weight than Δf_2 . This facilitates a reduction in the real-time frequency fluctuations in both inverter-side grids. To minimize the cost function J in Eq. (43), matrix $\mathbf{K}$ can be derived as $\mathbf{K} = \mathbf{R}^{-1} \cdot \mathbf{B}_R^T \cdot \mathbf{P}$, where $\mathbf{P}$ is a solution to the following equation:

$$\mathbf{A}_R^T \cdot \mathbf{P} + \mathbf{P} \cdot \mathbf{A}_R + \mathbf{Q} - \mathbf{P} \cdot \mathbf{B}_R \cdot \mathbf{R}^{-1} \cdot \mathbf{B}_R^T \cdot \mathbf{P} = \mathbf{O}. \quad (43)$$

Furthermore, all states in $\mathbf{X}_R(t)$ must be measured to calculate the optimal LQR, that is, $\mathbf{r}(t)$. As shown in Fig. 5, a Kalman filter can be employed to estimate the unknown states using the measured states or outputs of $\mathbf{Y}_R(t)$. The Kalman filter is a recursive estimator that combines predictions from the state–space model with measurements to provide an optimal estimate of the system states. The LQG controller can effectively estimate the unmeasured states using the Kalman filter, enabling the optimal control input $\mathbf{r}(t)$ to be calculated based on the complete state information.

The estimated value of $\mathbf{X}_R$ is denoted as $\hat{\mathbf{X}}_R$. Successful operation of the Kalman filter results in $\hat{\mathbf{X}}_R$ being close to $\mathbf{X}_R$, and the output $\mathbf{Y}_R(s)$ of the augmented state–space model can be calculated using the transfer function matrices $\mathbf{V}(s)$ and $\mathbf{w}(s)$ as follows:

$$\mathbf{Y}_R(s) = \mathbf{V}(s) \cdot \mathbf{w}(s) = \mathbf{C}_R [s\mathbf{I} - (\mathbf{A}_R - \mathbf{B}_R \mathbf{K})]^{-1} \mathbf{B}_R \cdot \mathbf{w}(s). \quad (44)$$

In Eqs. (44), $\mathbf{Y}_R(s)$ represents the transfer function matrix that relates the state variables to the outputs, and $\mathbf{w}(s)$ denotes the disturbance input to the system. By incorporating the estimated $\mathbf{X}_R$ value, the Kalman filter enables the system output $\mathbf{Y}_R(s)$ to be estimated based on the augmented state–space model.

4.2. Eigenvalue analysis of the proposed frequency scheme

An eigenvalue analysis was conducted to verify the stability of the proposed control scheme. Fig. 6 shows the eigenvalues of the closed-loop system that considers the variations in the system parameters and controller gain of the MT-HVDC system. As shown in Fig. 6(a)–(c), as the DC link capacitance C_{dc} , DC link reactance L_{dc} , and HVDC-linked equivalent inductance L_{eq} increased from 12 to 50 μF , 2.1×10^{-5} to 4.2×10^{-3} pu, and 0.01–0.30 pu, respectively, the complex conjugate eigenvalues moved closer to the imaginary axis. This implies an increase in the overshoot and settling time. However, all eigenvalues remained in the left-half plane, confirming that the proposed control scheme based on LQG ensures grid frequency stability, even in the various environments of the MT-HVDC linked grids. In contrast, Fig. 6(d) illustrates that the poles initially located on the left-hand side moved further from the

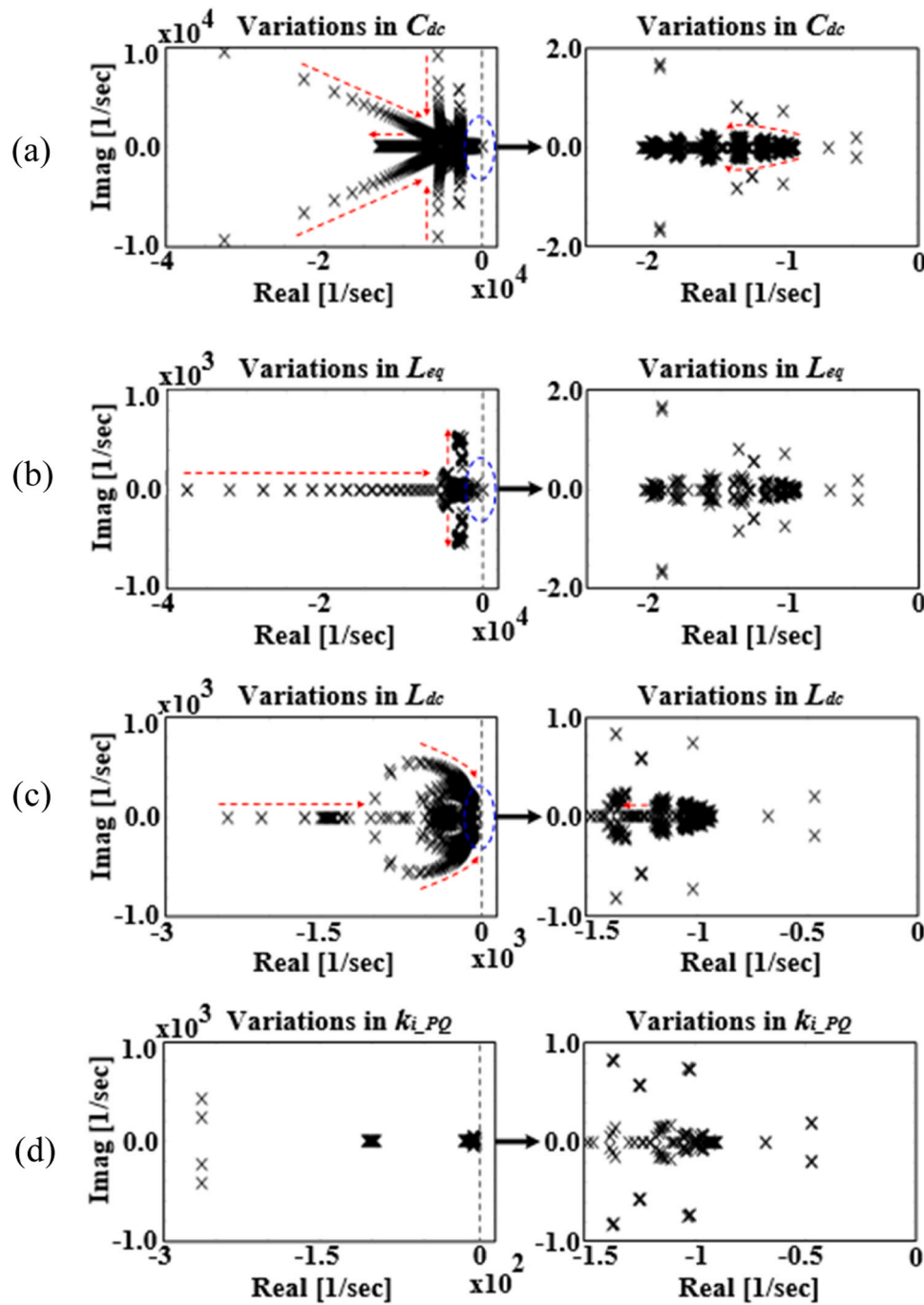


Fig. 6. Root loci of the proposed controller for increasing (a) C_{dc} from 12 to $50 \mu\text{F}$, (b) L_{dc} from 2.1×10^{-5} to 4.2×10^{-5} pu, (c) L_{eq} from 0.01 to 0.3 pu, and (d) k_{i_PQ} from 80 to 100 .

imaginary axis as k_{i_PQ} increased. This behavior suggests that the outer loop controller of VSC_n achieves a faster response time while maintaining a stable operation with an increasing value of k_{i_PQ} . In other words, Fig. 6(d) shows that the proposed control scheme is stable even when various control parameters are considered.

5. Case studies and results

5.1. Test system and simulation conditions

The proposed optimal control strategy was evaluated by utilizing the parameters and operational conditions of two HVDC systems: a general test case and a real project case. In the general test case, the parameters

and configuration provided in MATLAB example file were modified, and the inertia and damping of the governors and turbines in each grid were determined by referring to (Li et al., 2020; Fang et al., 2018; Ambia et al., 2022). In the actual project, the system was configured by adjusting the parameters within the context of the Zhangbei HVDC project, which resembles the general test case (Wang et al., 2019; Gao et al., 2020; Li et al., 2021). A dynamic model of the MT-HVDC linked grids was implemented using MATLAB/Simulink (Simulink). Each grid primarily comprised SGs, transmission lines, loads, and either VSC or VSC_n for interfacing with the MT-HVDC system. The MT-HVDC converters in Simulink were modeled using average circuit models, incorporating inner- and outer-loop feedback controllers in the d - q frame. The DC line between the MT-HVDC converters was represented using a

Table 1

Parameters for the case studies with the general test case and real project.

General Test Case				
Devices	Parameters	Values	Unit	
HVDC Converters	Rated Voltage	$V_{rated,ac}$	26	[kV]
		$V_{rated,dc}$	± 320	[kV]
	Rated Power	S_{rated}	600	[MVA]
DC networks	Phase reactance	R_{eq}, L_{eq}	0.005, 0.150	[pu]
	Outer loop gains	$k_{p,vdc}, k_{p,vdc}, k_{p,pq}, k_{i,pq}$	70, 200, 0.4, 80	[pu]
	Inner loop gains	$k_{p,dq}, k_{i,dq}$	0.4, 80	[pu]
	DC-link	R_{dc}	3.76	[Ω]
	parameter	L_{dc}	0.0189	[mH]
		C_{dc}	12	[μ F]
Grid	Inertia	M_r, M_1, M_2, M_3	10, 5, 10, 1	-
Governors and Turbines	Damping	D_r, D_1, D_2, D_3	1, 1, 1, 5	-
	Time constants	T_{gr}, T_{chn}	0.2, 0.5	[s]
PFC	Emulated inertia	T_{fi}, v_i	0.01, 1	[s]
	Droop gains (HVDC)	R_i	0.5	[pu]
	Droop gains (Gen.)	R_{gi}	0.5	[pu]
Control scheme (Conv.)	PI gains (HVDC)	KP_r, KP_i, KI_r, KI_i	0.8, 0.8, 1.5, 1.5	[pu]
	PI gains (Gen.)	$KP_{gr}, KP_{gi}, KI_{gr}, KI_{gi}$	0.8, 0.8, 1.5, 1.5	[pu]
Real Project				
Devices	Parameters	Values	Unit	
HVDC Converters	Rated Voltage (AC)	$V_{rated,ac}$	230, 525, 525, 230	[kV]
	Rated Voltage (DC)	$V_{rated,dc}$	± 500	[kV]
	Rated Power	$S_{rated} (r,1,2,3)$	1500, 1500, 3000, 3000	[MVA]
	Phase reactance	$R_{eq} (r,1), R_{eq} (2,3), L_{eq} (r,1), L_{eq} (2,3)$	0.0176, 0.0088, 0.04, 0.075	[pu]
DC networks	Resistance (per unit)	$R_{dc,unit}$	0.009735	[Ω /km]
	Inductance (per unit)	$L_{dc,unit}$	0.0176	[mH/km]
	Capacitance (per unit)	$C_{dc,unit}$	0.001367	[μ F/km]
	Line length	$l (r-1), (1-2), l (2-3), (3-r)$	207.1, 189.1, 208.4, 49.6	[km]
	Smoothing Reactor	L_{sm}	200	[mH]
Governors and Turbines	Inertia	M	5	-
	Damping	D	1	-

bi-pole model (Xiang et al., 2021) to achieve a higher modeling accuracy. The generators in the AC grids were assumed to be non-reheat turbine generators, and the aggregate dynamic responses were modeled using second-order transfer functions (Kundur, 1994). Table 1 presents a comprehensive list of the model parameters for the MT-HVDC linked grids and the corresponding control gains at the local level. The inertia and damping characteristics of the grids connected to the inverter were deliberately varied to assess the impact of the proposed algorithm. Especially, as shown in Table 1, a large smoothing reactor was installed a large smoothing reactor to suppress the increase rate of fault current in the the Zhangbei HVDC project. Fig. 7 illustrates the load demand variations (ΔP_n) for all inverter-side grids (n), which reflect the scaled-down regulation signals over a period of 250 s (Fast response regulation signal, PJM). Considering the operating conditions, the proposed frequency control scheme was analyzed and compared with conventional controllers. Table 2 lists the main features of the proposed scheme (Case 1) and the conventional schemes (Cases 2 and 3). In Case 2, PI controllers that could be locally operated were adopted to regulate the frequency deviations in the regional grid with variations of V_{dc}^{ref} and P_{gr}^{ref} . In Case 3, PI controllers that could be centrally operated were employed to minimize the FR of inverter-side grids with variations of V_{dc}^{ref} and P_{gr}^{ref} . The effects of SFC in MT-HVDC grids were analyzed by comparing the simulation results of Cases 1 and 2. A comparative analysis of the simulation results between Cases 1 and 3 confirmed the effect of optimal frequency control on the MT-HVDC grid. Importantly, in all three cases (Cases 1–3), IE and PFC were implemented in the feedback loops. Furthermore, the gains of PI controllers were modified for Cases 2 and 3, in addition to Case 1. This approach enabled fair comparisons between the proposed and conventional control schemes in terms of cooperative SFC for the active response of the converters in Cases 1–3.

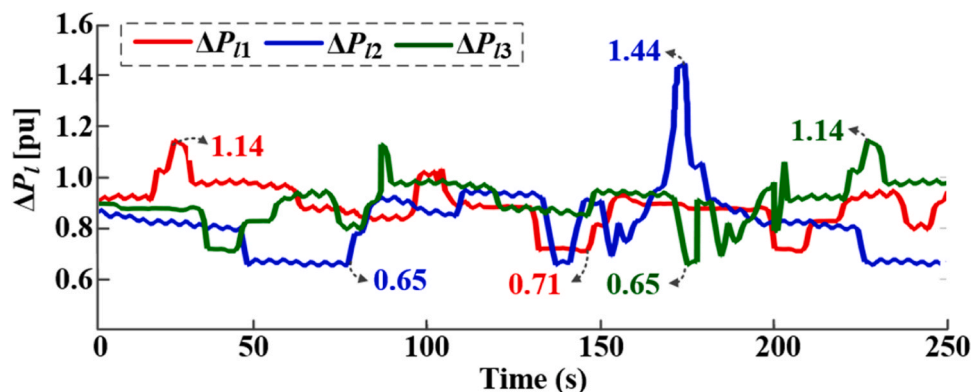
5.2. Validation of the state-space model for MT-HVDC grids

Fig. 8 presents a comparison between the responses of the state space and dynamic Simulink models of the MT-HVDC linked grids to a step increase in $\Delta P_{1-3} = 0.1$ pu at $t = 1.0$ and 1.1 s in the general test case and

Table 2

Features of FR strategies.

Mode		Reference value		Control Location	Optimal Gains	Control Scheme
		P_k^{ref}	V_{dc}^{ref}			
Pro.	Case 1	time-varying	time-varying	Central	O	LQG
Conv.	Case 2	time-varying	time-varying	Local	O	PI
	Case 3	time-varying	time-varying	Central	O	PI

**Fig. 7.** Load demand profile of Grid 1–3.

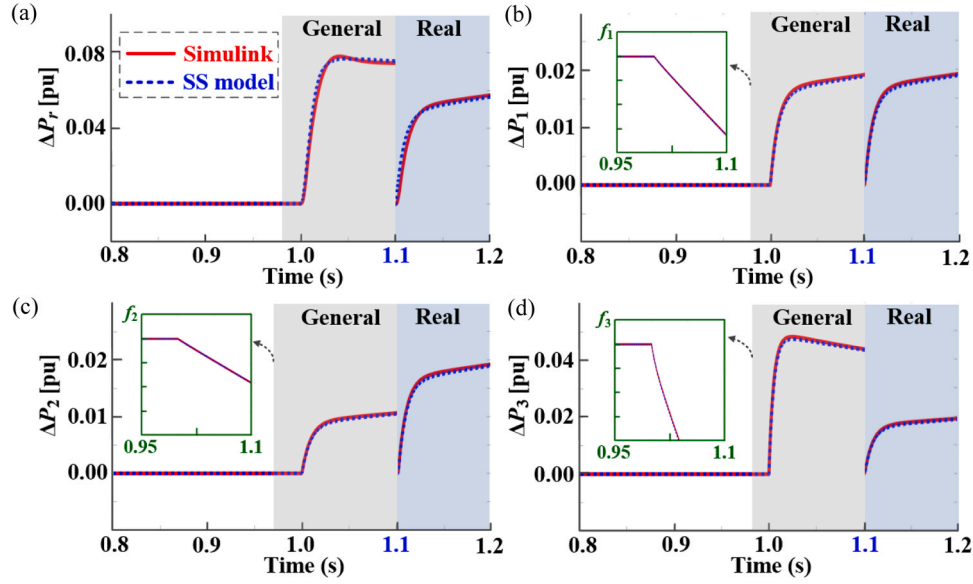


Fig. 8. Comparison between the state space model and Simulink model for the MT-HVDC linked grids when ΔP_{In} increased by 0.1 pu: (a) P_r , (b) $P_1(f_1)$, (c) $P_2(f_2)$, (d) $P_3(f_3)$.

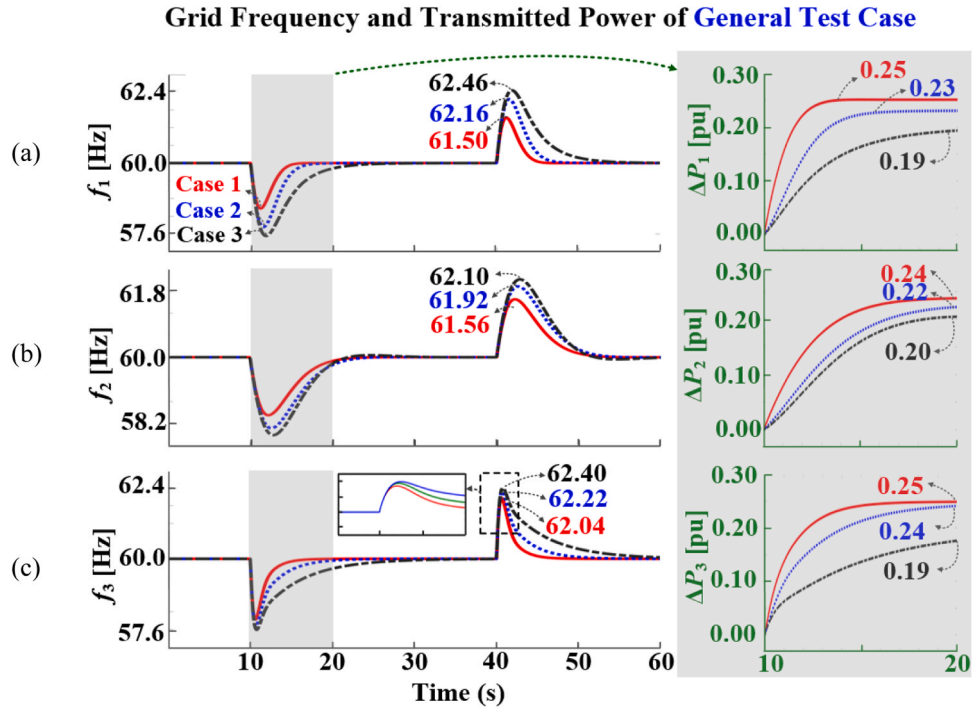


Fig. 9. Responses of the FR strategies to stepwise variations for general test case when ΔP_{In} increased in 0.3 pu: (a) f_1 , (b) f_2 , and (c) f_3 .

Table 3

Comparison of stepwise response test results in general test case.

Maximum Variations	Case 1		Case 2		Case 3	
	Individual	Total	Individual	Total	Individual	Total
$ \Delta f_1 _{max}$ [Hz]	0.025		0.036	0.105	0.041	0.116
$ \Delta f_2 _{max}$ [Hz]	0.026	0.085	0.032		0.035	
$ \Delta f_3 _{max}$ [Hz]	0.034		0.037		0.040	
Ratio [%]	-		19.1		26.8	

real project, respectively. Fig. 8 shows that the dynamic responses of the two models exhibit a high degree of similarity for each transient profile of ΔP_r and ΔP_{1-3} across all MT-HVDC terminals. Furthermore, in each ΔP profile, the grid frequency changes in the transient section can be observed. This indicates a strong consistency between the two models, even when the MT-HVDC linked grids experience relatively large variations in load demands under normal conditions, as depicted in Fig. 8 (a)–(d). These findings validate the accuracy of the simulation results presented in Sections 5.3, 5.4, and 5.5, which were obtained using the state space model of the MT-HVDC grids with the proposed controller. Specifically, in the general test case, the observations displayed in Fig. 8 (b)–(d) concern the distinct response time and frequency drop slope for

Table 4

Step response test results for generated power in the general test case.

Maximum Variations	Case 1	Case 2	Case 3
$ \Delta P_{g1} _{\max}$ [pu]	0.05	0.07	0.11
$ \Delta P_{g2} _{\max}$ [pu]	0.06	0.08	0.10
$ \Delta P_{g3} _{\max}$ [pu]	0.05	0.06	0.11

each profile during a transient situation. The disparity arises owing to inherent dissimilarities in the inverter-side grid characteristics, such as inertia and damping. This supports the coordinated FR of MT-HVDC linked grids.

5.3. Comparisons of stepwise load variations

Fig. 9 shows the stepwise responses of the MT-HVDC linked grids to load variations ΔP_{ln} in the general test case. The load variations increased by 0.3 pu for all grids n at $t = 10$ s and continued for a duration of 60 s. Fig. 9 (left side) shows f_{1-3} for these step responses. The corresponding numerical results are listed in Table 3. In the proposed scheme, referred to as Case 1, the frequency deviations in the inverter-side grids n are smaller than those in the conventional control scheme (Cases 2 and 3). Specifically, Case 1 resulted in a 19.1 % decrease in the sum of the maximum frequency deviations ($\sum_n |\Delta f_n|_{\max}$) compared to Case 2, and a 26.8 % decrease compared to Case 3, as indicated in Table 3. Furthermore, in Case 1, the frequencies f_n reverted to the nominal values more rapidly than in Cases 2 and 3. That is, the proposed control scheme based on the LQG controller was more effective at minimizing the frequency variations in MT-HVDC linked grids, even with different grid characteristics (M , D). Additionally, Fig. 9 (right side) illustrates the corresponding profiles of ΔP_n in load increase at 10 s. As shown in Fig. 9 (right side), Case 1 reaches a steady state faster than Cases 2 and 3 during the transient period. This improvement can be attributed to the optimal allocation of load demand variations to the VSCs and generators in both grids achieved using the proposed control scheme.

Specifically, both ΔP_1 and ΔP_2 , which were connected to Grids 1 and 2, increased by 0.25 and 0.24 pu, while ΔP_3 connected to Grid 3 increased by 0.25 pu at $t = 10$ s. Meanwhile, ΔP_{g1} , ΔP_{g2} , and ΔP_{g3}

increased by 0.05, 0.06, and 0.05 pu, respectively, as indicated in Table 4. The proposed scheme reduces frequency fluctuations by utilizing VSCs more effectively than the conventional scheme. The comparison results demonstrate that the enhanced performance of Case 1 can be attributed to the optimal power sharing among the MT-HVDC linked grids through the DC transmission lines, both in the transient and steady states. This improved performance was achieved without requiring an additional power supply or excessive variations in the generated and transmitted power. Fig. 10 illustrates the profiles of Δf_{1-3} and ΔP_{1-3} , corresponding to the modified parameters of the Zhangbei HVDC project. In Case 1, an increase of 0.2 pu in ΔP_{ln} led to a proportional increase of 0.16 pu in ΔP_{1-3} . This indicates that the LQG controller effectively mitigates frequency deviations by redistributing load demand fluctuations across the generators on each grid. This approach facilitates the optimized regulation of the DC-link voltage and frequency deviation, resulting in enhanced power-generation performance under both transient conditions. Conversely, the application of PI controllers in Cases 2 and 3 compelled the generators to compensate for load changes, which were generally concentrated on the same side of the HVDC system. Consequently, the transient period showed only a marginal contribution from the HVDC system to FR on the inverter side. This limitation, in turn, results in a pronounced frequency variation in f_{1-3} . This underscores the efficiency of strategies such as IE and droop control, which leverage the rapid responsiveness of HVDC converters. Notably, the outcomes of Case 1 yielded a remarkable reduction of

Table 5

Comparison of stepwise response test results in the real project.

Maximum Variations	Case 1		Case 2		Case 3	
	Individual	Total	Individual	Total	Individual	Total
$ \Delta f_1 _{\max}$ [Hz]	0.017		0.024	0.073	0.027	0.083
$ \Delta f_2 _{\max}$ [Hz]	0.022	0.057	0.025		0.028	
$ \Delta f_3 _{\max}$ [Hz]	0.018		0.024		0.028	
Ratio [%]	-		21.9		31.3	

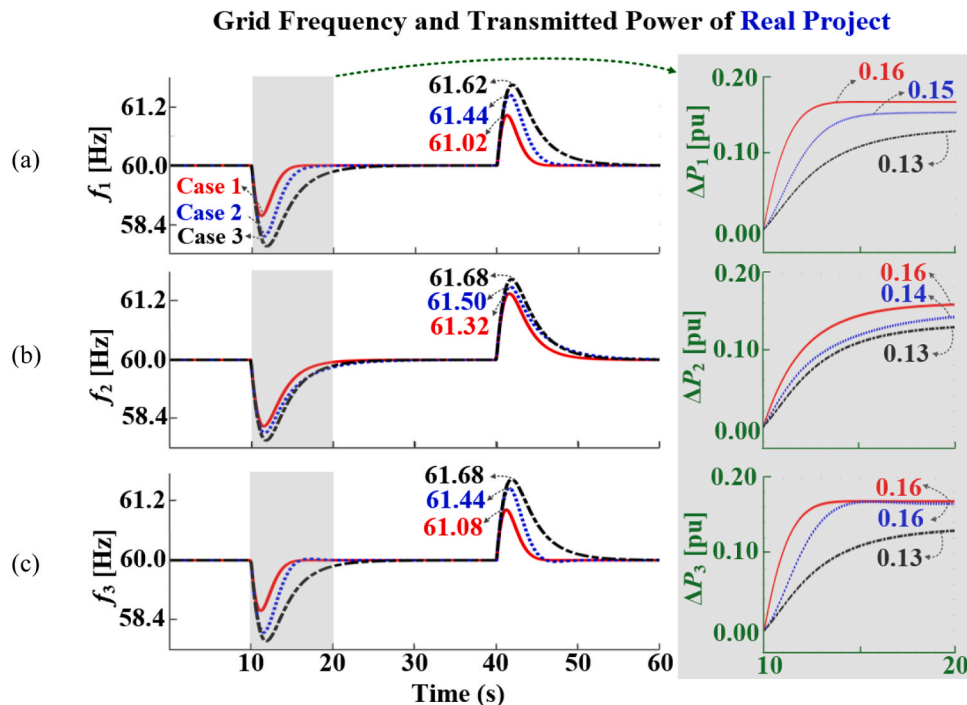


Fig. 10. Responses of the FR strategies to stepwise variations for real project when ΔP_{ln} increased in 0.2 pu: (a) f_1 , (b) f_2 , and (c) f_3 .

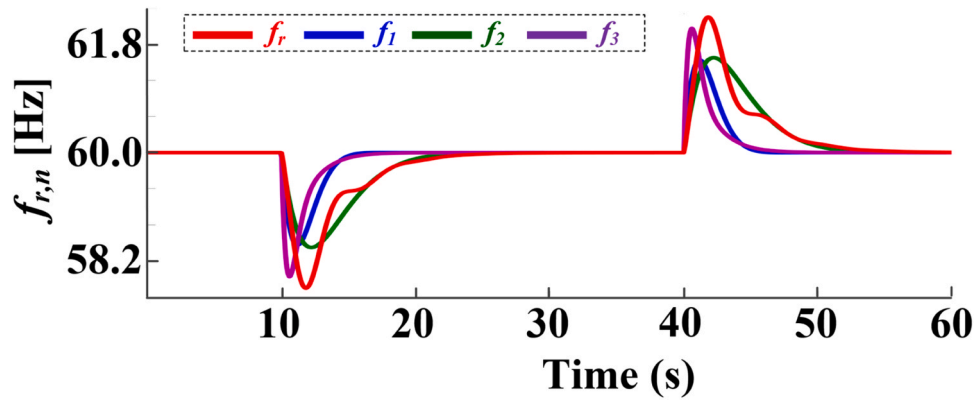


Fig. 11. Comparison with the frequency response of all grids to stepwise variations for test case when ΔP_{ln} and ΔP_{lr} increased in 0.3 pu under conditions with different grid characteristics.

Table 6

Comparison of stepwise response test results in general test case.

Grid frequency	Main Grid	Grid1	Grid2	Grid3
$f_{r,n}(n=1,2,3)$ [Hz]	62.28	61.50	61.56	62.04
Ratio [%]	-	1.27	1.17	0.37

※ Ratio [%] = $(f_r - f_n) / f_n \times 100$ [%]

21.9 % in the sum of the maximum frequency deviations when compared to Case 2 and an even more substantial decrease of 31.3 % relative to Case 3. These results, as documented in Table 5, robustly substantiate the frequency-control efficacy of the proposed LQG controller in a real project with a large smoothing reactor installed on the DC line. In Fig. 11, the frequencies of each grid including main grid are compared when the load within all grids including main grid are increased equally to 0.3 pu. The frequency response during stepwise

Grid Frequency of General Test Case

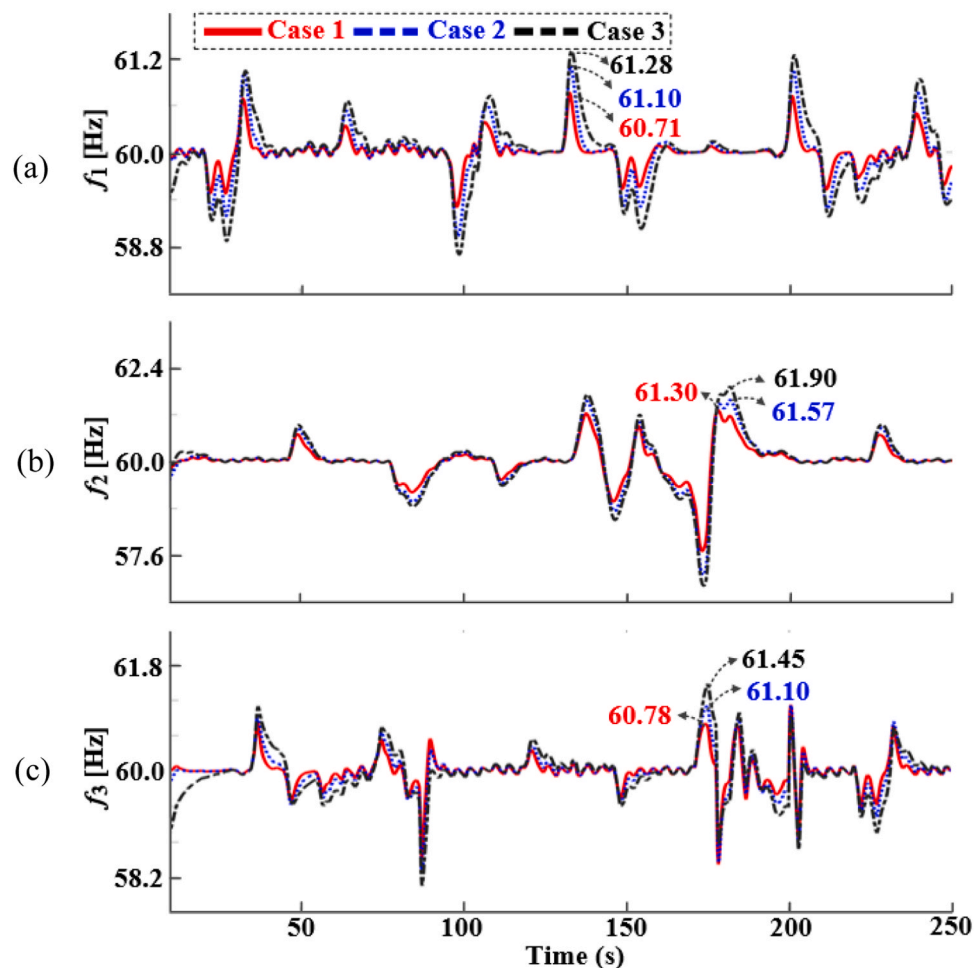


Fig. 12. Responses of the FR strategies to continuous variations in ΔP_i : (a) f_1 , (b) f_2 , and (c) f_3 .

Table 7

Comparison of the continuous response test results in the general test case.

RMS Variations	Case 1		Case 2		Case 3	
	Individual	Total	Individual	Total	Individual	Total
$\Delta f_{1,rms}$ [Hz]	0.007	0.024	0.011	0.036	0.016	0.047
$\Delta f_{2,rms}$ [Hz]	0.011		0.016		0.018	
$\Delta f_{3,rms}$ [Hz]	0.006		0.009		0.013	
Ratio [%]	-		33.3		48.9	

load fluctuations was evaluated by introducing SFC into the main grid. In this case, PI Controller was used as the SFC applied to the main grid. Table 6 presents the maximum value of grid frequency including main grid. In Table 6, it can be seen that the maximum value of the frequencies for each grid increased by 1.27 %, 1.17 %, and 0.37 % when compared to Grid 1, Grid 2, and Grid 3, respectively. This is because a lot of power is supplied from the main grid to other grids to minimize frequency fluctuations in the grid linked to the inverter. However, in this paper, it is assumed that the main grid is a robust system with a very large M and that it supplies sufficient power. Note that even if the load increases to the same value, the minimum frequency values of the grids connected to the inverter are different. In this paper, since the grid characteristics (M , D) of the general test case presented in Table 1 are different, the frequency fluctuations of each grid are different even if the same size load is applied. One of the main contributions in this paper is that the proposed frequency optimal control algorithm minimizes frequency fluctuations even when applied to MT-HVDC composed of grids with different characteristics. Therefore, when the proposed frequency optimal control algorithm was applied to an MT-HVDC system with different characteristics of each grid under the assumption that the main grid is a robust system with a very large M and supplies sufficient power, it was confirmed that the frequency variation of the grid linked to the inverter was minimized during stepwise load variation.

5.4. Comparisons of Continuous Load Variations

The proposed frequency control scheme was evaluated by conducting tests considering continuous variations in the load demand ΔP_{ln} over a time period of 250 s in the general test case, as shown in Fig. 12. Table 7 presents the numerical results of the RMS values, calculated as $\{\sum_j \Delta f_{n,j}^2 / J\}^{1/2}$ and $\{\sum_j \Delta P_{n,j}^2 / U\}^{1/2}$, where j denotes the index of the measurement sample and J denotes the total number of samples. When comparing the different cases, we observed that Case 1 resulted in significantly smaller variations in frequencies for the inverter-side grids than Cases 2 and 3. Specifically, the sum of $\Delta f_{1-3,rms}$ in Case 1 was 33.3 % and 48.9 % smaller than those in Cases 2 and 3, respectively. In Fig. 13, the result of the DC voltage variation (ΔV_{dcr}) was presented when

a continuous various simulation was performed within the load demand ΔP_{ln} for 250 seconds. As can be seen in Eq. (38), ΔV_{dcr}^{ref} is one of the inputs of the proposed LQG controller. In proposed strategy, the integration was aimed at minimizing the cumulative variations in Δf_n and ΔV_{dcr} , while ensuring that the recovered value equaled the nominal values in steady state. In other words, the change in ΔV_{dcr} is small because the variation in ΔV_{dcr}^{ref} is minimized by the LQG controller. Table 8 reveals analogous outcomes between the continuous variation applied to the real project. Notably, the cumulative $\Delta f_{1-3,rms}$ in Case 1 exhibited a reduction of 32.4 % and 47.7 % compared to Cases 2 and 3, respectively. This finding underscores the consistent efficacy of the proposed approach in minimizing frequency deviations within a real project. This indicates that the proposed scheme effectively enhances the FR of MT-HVDC linked grids, while simultaneously reducing power variations through optimal and coordinated sharing of the time-varying power outputs of the VSCs and SGs. Case studies confirm that the proposed LQG controllers effectively enhances the FR of MT-HVDC linked grids consistently.

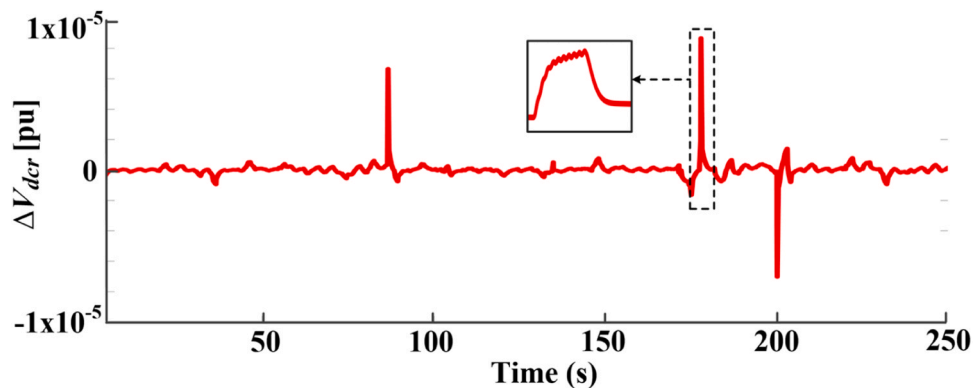
5.5. Effect of communication time delays

Several case studies were conducted to investigate the effect of communication time delays on the proposed scheme. Communication time delays in large power systems generally range from tens to hundreds of milliseconds (El Moursi et al., 2008). Fig. 14 presents a comparison between the step responses of f_1 , f_2 , and f_3 in Cases 1 and 2, considering variations in ΔP_{11} of 0.3 pu, while gradually increasing t_d from 0 to 0.15 s in the general test case. As t_d increased, Case 2 exhibited larger $|\Delta f_n|_{max}$ values than Case 1. In Grid 3, the difference in the $|\Delta f_3|_{max}$ values between Cases 1 and 2 was insignificant compared to other grids. This is because Grid 3 was a weak grid with a large damping value. Additionally, Case 1 ensured stable and smaller variations in the intermittent grid frequency for all t_d values up to 0.15 s, as depicted in Table 9. The results demonstrate that the proposed scheme enhances the performance effectiveness and robustness of the FR in MT-HVDC linked grids, even in the presence of relatively large communication time delays.

Table 8

Comparison of the continuous response test results in the real project.

RMS Variations	Case 1		Case 2		Case 3	
	Individual	Total	Individual	Total	Individual	Total
$\Delta f_{1,rms}$ [Hz]	0.006	0.023	0.011	0.034	0.015	0.044
$\Delta f_{2,rms}$ [Hz]	0.011		0.015		0.017	
$\Delta f_{3,rms}$ [Hz]	0.006		0.008		0.012	
Ratio [%]	-		32.4		47.7	

**Fig. 13.** DC voltage variation of main grid to continuous variations in ΔP_{ln} .

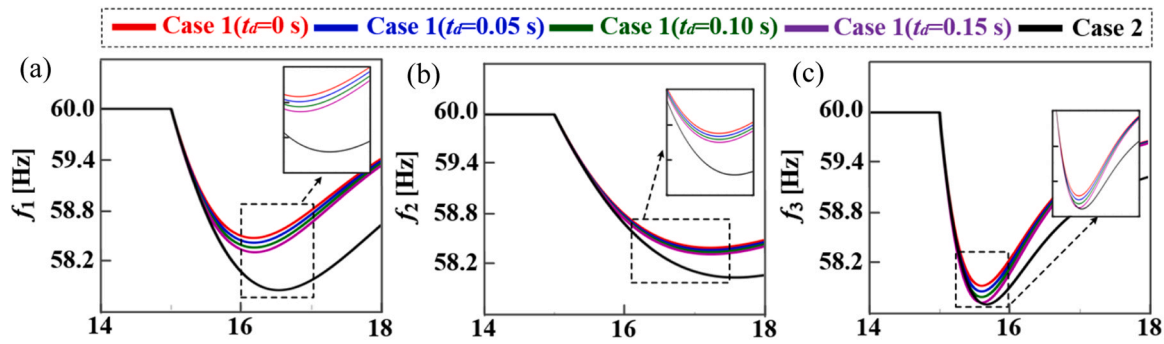


Fig. 14. Grid frequency of the proposed (Case 1) and conventional (Case 2) strategies when t_d ranges from 0.0 s to 0.15 s in general test case: (a) f_1 , (b) f_2 , and (c) f_3 .

Table 9

Comparison of the time delay test results in the general test case.

	t_d [ms]	$ \Delta f_1 _{\max}$ [Hz]	$ \Delta f_2 _{\max}$ [Hz]	$ \Delta f_3 _{\max}$ [Hz]
Case 1	50	0.026	0.027	0.035
	100	0.028	0.027	0.036
	150	0.029	0.028	0.038
Case 2		0.036	0.032	0.038

5.6. Effect of the other types of HVDC configuration

Fig. 15 displays a schematic of an MT-HVDC grid comprising four VSCs and three grids. Each VSC is connected to the PCC of each grid via a high-voltage feeder. In particular, both VSC1 and VSC2 are connected to a Grid1 for multi-infeed configuration. This configuration is similar to Jeju Island, in which two HVDC systems provide active power from the mainland. The simulation results are detailed in Fig. 15 and Table 9. Fig. 14 illustrates the step response profiles of f_{1-2} and ΔP_{1-3} for the MT-HVDC grids when both ΔP_{11} and ΔP_{12} are increased by 0.3 pu at $t = 10$ s and continued for a duration of 60 s. As listed in Table 10, the proposed strategy decreases $|\Delta f_1|_{\max} + |\Delta f_2|_{\max}$ by 36.1 % and 43.5 %, respectively, compared to the conventional strategies (i.e., Cases 2 and 3, respectively). These findings, as seen in Table 10, indicate that the proposed controller effectively mitigates frequency deviations in the grid, establishing its applicability to diverse HVDC configurations

6. Conclusion

This paper presents a novel scheme for achieving FR in MT-HVDC linked grids through power-sharing between VSC_n and SGs. The proposed approach involves the development of a state space model for MT-HVDC linked grids that integrates IE and droop controllers. The accuracy of the developed state space model was verified by comparing it with a comprehensive MATLAB/Simulink model. To bolster FR within the MT-HVDC grids, we designed an advanced LQG controller. The designed LQG controller optimizes the power allocation between VSC_n and SGs. The proposed scheme are evaluated through meticulous

sensitivity and eigenvalue analyses, considering inter-grid measurements and system parameters. The results of comparative case studies, encompassing both a general test case and real project with modified parameters of the Zhengbei HVDC system, unequivocally demonstrated that the proposed scheme outperforms the conventional scheme in reducing frequency deviations across all MT-HVDC linked grids. Moreover, compared with the conventional PI control scheme, the proposed scheme exhibited improved effectiveness and robustness in achieving optimal FR for MT-HVDC linked grids under various operating conditions. Additionally, the proposed scheme effectively mitigated grid frequency fluctuations, even in the presence of communication time delays between the proposed and local controllers of the VSC_n and RVSC. In summary, our proposed scheme marks a significant leap forward in the realm of optimal FR for MT-HVDC linked grids. The advancement is underscored by comprehensive analyses and comparative studies that confirm its effectiveness and superiority. In this paper, an optimal frequency control strategy is proposed under the assumption that the main grid is very large. Future work will focus on validating the algorithm proposed in this paper when all grids are identical.

CRediT authorship contribution statement

Seung-Ill Moon: Validation, Methodology. **Yong-Tae Yoon:** Validation, Conceptualization. **Do-Hoon Kwon:** Writing – review & editing, Supervision, Conceptualization. **Jae-Sung Lim:** Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization.

Table 10

Additional comparison of stepwise response test results in MT-HVDC.

Maximum Variations	Case 1		Case 2		Case 3	
	Individual	Total	Individual	Total	Individual	Total
$ \Delta f_1 _{\max}$ [Hz]	0.016	0.039	0.024	0.061	0.028	0.069
$ \Delta f_2 _{\max}$ [Hz]	0.023		0.037		0.041	
Ratio [%]	-		36.1		43.5	

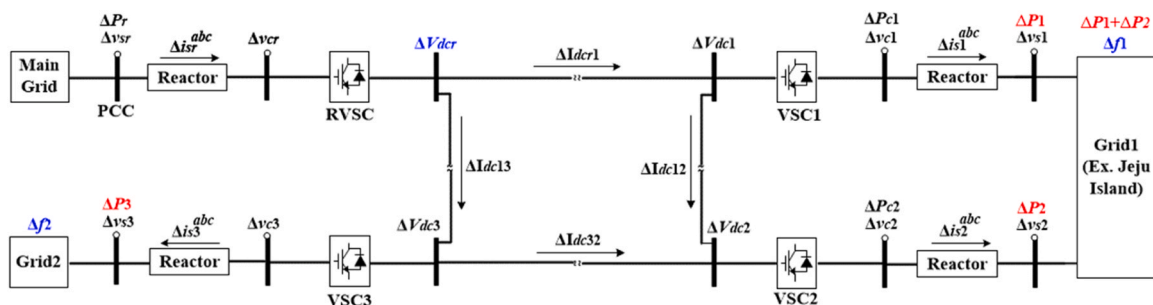


Fig. 15. Schematic diagram of MT-HVDC grids added to verify the effectiveness of the proposed control strategy.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Appendix A

In (1)–(3), the coefficient matrices are presented using the parameter of grids and MT-HVDC systems as follows:

$$\mathbf{A}_r = \begin{bmatrix} a_{1(1,1)} & \cdot & \cdot & \cdot & \cdot & \cdot & a_{1(7,7)} \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & a_{1(4,4)} & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{1(7,1)} & \cdot & \cdot & \cdot & \cdot & \cdot & a_{1(7,7)} \end{bmatrix}, \mathbf{A}_{1-3} = \begin{bmatrix} a_{2(1,1)} & \cdot & \cdot & \cdot & \cdot & \cdot & a_{2(7,7)} \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & a_{2(4,4)} & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{2(7,1)} & \cdot & \cdot & \cdot & \cdot & \cdot & a_{2(7,7)} \end{bmatrix}, \quad (\text{A1})$$

$$a_{1(1,1)} = (-R_{eq} - k_{p_dq})/L_{eq}, a_{1(1,3)} = (-k_{p_dq} \cdot k_{p_Vdc})/L_{eq}, a_{1,2(1,4),(2,5)} = k_{p_dq}/L_{eq},$$

$$a_{1,2(1,6),(2,7)} = 1/L_{eq}, a_{1(2,2)} = (-R_{eq} - k_{p_dq} + k_{p_dq} \cdot k_{p_PQ} \cdot v_{sr}^q)/L_{eq},$$

$$a_{2(1,1)} = (-R_{eq} - k_{p_dq} - k_{p_dq} \cdot k_{p_PQ} \cdot v_{sn}^d)/L_{eq}, a_{2(2,2)} = (-R_{eq} - k_{p_dq} + k_{p_dq} \cdot k_{p_PQ} \cdot v_{sn}^q)/L_{eq},$$

$$a_{2(3,1)} = (i_{sn}^d \cdot v_{sn}^d \cdot k_{p_dq} \cdot k_{p_PQ} + i_{sn}^d \cdot k_{p_dq} + i_{sn}^q \cdot w \cdot L_{eq})/(C_{dcn} \cdot V_{dcn}),$$

$$a_{2(3,2)} = (v_{cn}^q + i_{sn}^d \cdot w \cdot L_{eq} - k_{p_dq} \cdot k_{p_PQ} \cdot i_{sr}^q \cdot v_{sr}^q + k_{p_dq} \cdot i_{sn}^q)/(C_{dcn} \cdot V_{dcn}),$$

$$a_{2(3,3)} = (-i_{sn}^d \cdot v_{cn}^d - i_{sn}^q \cdot v_{cn}^q)/(C_{dcn} \cdot V_{dcn}^2), a_{2(3,4)} = (-i_{sn}^d \cdot k_{p_dq})/(C_{dcn} \cdot V_{dcn}),$$

$$a_{2(3,5)} = (-i_{sn}^q \cdot k_{p_dq})/(C_{dcn} \cdot V_{dcn}), a_{2(3,6)} = (-i_{sn}^d)/(C_{dcn} \cdot V_{dcn}), a_{2(3,7)} = (-i_{sn}^q)/(C_{dcn} \cdot V_{dcn}),$$

$$a_{2(4,1)} = -k_{i_PQ} \cdot v_{sn}^d,$$

$$a_{2(5,2)} = k_{i_PQ} \cdot v_{sn}^q,$$

$$a_{2(6,1)} = -k_{i_dq} - k_{i_dq} \cdot k_{p_PQ} \cdot v_{sn}^d, a_{2(6,4)} = k_{i_dq},$$

$$a_{2(7,2)} = -k_{i_dq} + k_{p_PQ} \cdot k_{i_dq} \cdot v_{sn}^q, a_{2(7,5)} = k_{i_dq}$$

$$\mathbf{B}_r = \begin{bmatrix} b_{1(1,1)} & \cdot \\ \cdot & b_{1(2,2)} \\ b_{1(3,1)} & b_{1(3,2)} \\ b_{1(4,1)} & \cdot \\ \cdot & b_{1(5,2)} \\ b_{1(6,1)} & \cdot \\ \cdot & b_{1(7,2)} \end{bmatrix}, \mathbf{B}_{1-3} = \begin{bmatrix} \cdot & \cdot \\ \cdot & b_{2(2,2)} \\ \cdot & \cdot \\ \cdot & \cdot \\ \cdot & b_{2(5,2)} \\ \cdot & \cdot \\ \cdot & b_{2(7,2)} \end{bmatrix}, \quad (\text{2})$$

$$b_{1(1,1)} = (k_{p_dq} \cdot k_{p_Vdc})/L_{eq}, b_{1(2,2)} = (k_{p_dq} \cdot k_{p_PQ})/L_{eq},$$

$$b_{1(3,1)} = (-i_{sr}^d \cdot k_{p_dq} \cdot k_{p_Vdc})/(C_{dcr} \cdot V_{dcr}), b_{1(3,2)} = (-i_{sr}^q \cdot k_{p_dq} \cdot k_{p_PQ})/(C_{dcr} \cdot V_{dcr}),$$

$$b_{1(4,1)} = k_{i_Vdc}, b_{2(5,2)} = k_{i_PQ},$$

$$b_{1(6,1)} = k_{i_dq} \cdot k_{p_Vdc}, b_{1(7,2)} = k_{i_dq} \cdot k_{p_PQ},$$

$$b_{2(2,2)} = (k_{p_dq} \cdot k_{p_PQ})/L_{eq}, b_{2(5,2)} = k_{i_PQ}, b_{2(7,2)} = k_{p_PQ} \cdot k_{i_dq}$$

$$\mathbf{C}_r = \mathbf{C}_{1-3} = \mathbf{I}_{7 \times 7} \quad (\text{A3})$$

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